Plectic action on cohomology of Hilbert modular varieties

洛 略

<u>Introduction</u> modular curve

$K \subset GL_2(\mathbb{A}_f) \rightsquigarrow Y_K/\mathbb{Q} = \{\text{elliptic curve} + \text{level } K\}$

$Y_K(\mathbb{C}) = GL_2(\mathbb{Q}) \backslash (\mathbb{C} \setminus \mathbb{R}) \times GL_2(\mathbb{A}_f)/K = \coprod \Gamma_i \backslash \mathcal{H}$

$H^1_p(Y_K) := \mathrm{Im}\left(H^1_c(Y_{K,\bar{\mathbb{Q}}}, \bar{\mathbb{Q}}_p) \to H^1(Y_{K,\bar{\mathbb{Q}}}, \bar{\mathbb{Q}}_p)\right) \circlearrowleft G_{\mathbb{Q}} = \mathrm{Gal}(\bar{\mathbb{Q}}|\mathbb{Q})$

$$H^1_p = \varinjlim_{K \subset GL_2(\mathbb{A}_f)} H^1_p(Y_K) = \bigoplus \pi^\infty \otimes \rho_\pi \quad , \dim \rho_\pi = 2$$

$\circlearrowleft$ $GL_2(\mathbb{A}_f) \times G_{\mathbb{Q}}$; $\pi = \pi^\infty \otimes \pi_\infty$ $\updownarrow$ newform of wt 2; $\pi^\infty \circlearrowleft GL_2(\mathbb{A}_f)$, $\rho_\pi \circlearrowleft G_{\mathbb{Q}}$

ρ_π^{ss} is determined by π.

<u>Fact</u>. ρ_π^{ss} is <u>irreducible</u> $\Rightarrow \rho_\pi = \rho_\pi^{ss}$.

<u>Cor</u>. $H^1_p(Y_K)$ is a <u>semisimple</u> $G_{\mathbb{Q}}$-representation

$k \geq 0 \rightsquigarrow GL_2 \curvearrowright \mathrm{Sym}^k \mathrm{Std} \longrightarrow \mathbb{L}_k$: p-adic local system on Y_K.

<u>Rem</u>. Same holds for $H^1_p(Y_K, \mathbb{L}_k)$

$F|\mathbb{Q}$ totally real, $d = [F:\mathbb{Q}]$, $F \otimes_{\mathbb{Q}} \mathbb{R} \simeq \mathbb{R}^d$

$K \subset GL_2(\mathbb{A}_f \otimes_{\mathbb{Q}} F) \rightsquigarrow$ Hilbert modular variety $Sh_K/\mathbb{Q}$.

$Sh_K(\mathbb{C}) \simeq GL_2(F) \backslash (\mathbb{C} \setminus \mathbb{R})^d \times GL_2(\mathbb{A}_f \otimes_{\mathbb{Q}} F)/K$

Sh_K : d-dim'l alg. var. $/\mathbb{Q}$

weight $\underline{k} = (k_1, \cdots, k_d ; w)$ $\quad k_1, \cdots, k_d \geq 0, \quad k_1 \equiv \cdots \equiv k_d \equiv w \pmod 2$

$\rightsquigarrow \mathbb{L}_{\underline{k}}$ = p-adic loc. sys. on Sh_K.

$$H^d_p := \varinjlim_{K} \mathrm{Im}\left(H^d_c(Sh_{K,\bar{\mathbb{Q}}}, \mathbb{L}_{\underline{k}}) \to H^d(Sh_{K,\bar{\mathbb{Q}}}, \mathbb{L}_{\underline{k}})\right)$$

$$\circlearrowleft$$

$$GL_2(\mathbb{A}_f \otimes F) \times G_{\mathbb{Q}} \qquad = H^d_{cusp} \oplus H^d_{non-cusp}$$

$$H^d_{cusp} = \bigoplus_{\pi = \pi^\infty \otimes \pi_\infty} \pi^\infty \otimes \underset{\underset{G_{\mathbb{Q}}}{\circlearrowleft}}{V(\pi)} \quad , \quad \dim V(\pi) = 2^d$$

$$\pi \rightsquigarrow \rho_\pi : G_F \longrightarrow GL_2(\bar{\mathbb{Q}}_p) \quad \text{irreducible}$$

<u>Fact</u>. $V(\pi)^{ss} \simeq \otimes\text{-}\mathrm{Ind}\, \rho_\pi$ <u>tensor induction of ρ_π</u>

Assume $F|\mathbb{Q}$ Galois, $\sigma \in \mathrm{Gal}(F|\mathbb{Q})$ $\rightsquigarrow$ $\rho_\pi^\sigma : G_F \longrightarrow GL_2(\bar{\mathbb{Q}}_p)$

$\tilde{\sigma} \in G_{\mathbb{Q}}$ (with $\tilde\sigma \mapsto \sigma$, $G_{\mathbb{Q}} \twoheadrightarrow \mathrm{Gal}(F|\mathbb{Q})$) $\qquad$ conjugate of ρ_π by $\tilde{\sigma}$

$$\otimes\text{-}\mathrm{Ind}\, \rho_\pi = \bigotimes_{\sigma \in \mathrm{Gal}(F|\mathbb{Q})} \rho_\pi^\sigma \quad , \quad \dim = 2^d$$

<u>Rem</u>. If $\dim \rho_\pi = 1$ $\rightsquigarrow$ transfer in CFT.

<u>Problem</u>. $\otimes\text{-}\mathrm{Ind}\, \rho_\pi$ reducible in general.

<u>Th</u> (Nekovář) $H^*_p(\mathbb{L}_{\underline{k}})$ is a <u>s.s. $G_{\mathbb{Q}}$-rep</u>.

<u>Cor</u> $V(\pi) = V(\pi)^{ss} \simeq \otimes\text{-}\mathrm{Ind}\, \rho_\pi$

<u>Main Th</u> (Jiang - P.) $V(\pi) \simeq \otimes\text{-}\mathrm{Ind}\, \rho_\pi$ if π is <u>not CM</u>

$(\Rightarrow \rho_\pi \not\simeq \mathrm{Ind}\, \chi$

$\Leftrightarrow \mathrm{Lie}\, \mathrm{Im}(\rho_\pi) \supset sl_2)$

<u>Rem</u>. We also have a version for completed cohomology.

Plectic structure. $G_{\mathbb{Q}} := \mathrm{Aut}(\bar{\mathbb{Q}}|\mathbb{Q})$

$$G_F^d \rtimes \mathfrak{S}_d \simeq G_F^{plec} := \mathrm{Aut}(\bar{\mathbb{Q}} \otimes_{\mathbb{Q}} F \,|\, F) \supset G_{\mathbb{Q}}$$

plectic Galois group

$$\bar{\mathbb{Q}} \otimes_{\mathbb{Q}} F \simeq \prod_{\tau : F \hookrightarrow \bar{\mathbb{Q}}} \bar{\mathbb{Q}} \quad \cup \quad G_F^d \rtimes \mathfrak{S}_d$$

Conj. (Nekovář - Scholl) $G_{\mathbb{Q}}$-action on $H^*(Sh_K, \bar{\mathbb{Q}}, \mathbb{L}_{\underline{k}})$ can be ext. to an action of G_F^{plec}.

ρ_π plectic induction $\underbrace{\mathrm{Ind}^{plec}\rho_\pi}_{\text{irreducible}} : P_F^{plec} \longrightarrow GL_{2^d}(\bar{\mathbb{Q}}_p)$ s.t. $\mathrm{Ind}^{plec}\rho_\pi|_{G_{\mathbb{Q}}} = \otimes\text{-}\mathrm{Ind}\,\rho_\pi$

Nekovář: construct local plectic group action. partial Frobenii at $\ell \neq p$

Jiang - Pi: construct "$\mathrm{Lie}\, G_{F_p}^{plec}$" partial Sen operator.

Th. (Sen) $L \subset \bar{\mathbb{Q}}_p$, $L|\mathbb{Q}_p$ finite, $\rho : G_L \longrightarrow GL_d(\mathbb{Q}_p)$, $\mathbb{C}_p = \widehat{\bar{\mathbb{Q}}_p}$

Sen constructed θ (Sen operator) $\in \mathfrak{gl}_d(\mathbb{C}_p)$

(1) $\theta \in \mathrm{Lie}\,\rho(I_L) \otimes \mathbb{C}_p$

(2) If $V \subset \mathfrak{gl}_d(\mathbb{Q}_p)$ $\mathbb{Q}_p$-subspace s.t. $V \otimes \mathbb{C}_p \ni \theta$, then $V \supset \mathrm{Lie}\,\rho(I_L)$

Construction of partial Sen operators

- $F = \mathbb{Q}$, P.
- $d > 1$, Rodriguez Camargo